

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.10 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Q.1 If A_m is the area bounded by $y = mx$ and $y = x^2$, then $\lim_{n \rightarrow \infty} \frac{\sum_{m=1}^n A_m}{n^4}$ is equal to

- (A) $\frac{1}{24}$ (B) 24 (C) 6 (D) $\frac{1}{6}$

Q.2 If $\lim_{n \rightarrow \infty} \left(\frac{1^{2013} + 2^{2013} + \dots + n^{2013}}{n^{2014}} \right)$ equals λ , then $\frac{1}{\lambda}$ equals

- (A) 2012 (B) 2011 (C) 2013 (D) 2014

Q.3 If $f(x) = e^{x^3 - 3x}$ then $\int_{-\frac{1}{2}}^{\frac{1}{2}} (\cot^{-1} f(x)) dx - \int_{-\frac{1}{2}}^{\frac{1}{2}} (\tan^{-1} f(-x)) dx$ equals

- (A) 0 (B) 1 (C) $\frac{1}{2}$ (D) $\ln \sqrt{e}$

Q.4 Consider, $f(x) = \frac{3x-2}{x+1}$. If $I_n = \int_{-n}^n f(\{x\}) dx = 8 \ln \left(\frac{e^3}{32} \right)$, $n \in I$ then n is

- (A) 1 (B) 2 (C) 3 (D) 4

Q.5 Let f be a function satisfying the relation $f\left(\frac{x}{x-1}\right) + f\left(\frac{1}{1-x}\right) = 4$. The value of $\int_{-\frac{3}{2}}^{\frac{5}{2}} f(x) dx$ is

- (A) 4 (B) 8 (C) 16 (D) 32

Q.6 If $\lim_{x \rightarrow 0} \frac{\int_0^x \sin t dt + \int_0^x \sin 2t dt + \dots + \int_0^x \sin pt dt}{x^2} = 5$, where $p \in N$, then p is

- (A) 1 (B) 2 (C) 3 (D) 4

- Q.7 If f is a continuous positive valued function such that $\int_0^1 xf(4x)dx + \int_0^2 t f(2t)dt = \frac{5}{\lambda} \int_0^4 z f(z) dz$, then value of λ is equal to
 (A) 1 (B) 2 (C) 4 (D) 16

- Q.8 Let $\frac{d}{dx}(F(x)) = \frac{e^{\sin x}}{x}$, $x > 0$. If $\int_1^4 \frac{3}{x} e^{\sin(x^3)} dx = F(K) - F(1)$, then one of the possible value of k , is
 (A) 15 (B) 16 (C) 63 (D) 64

- Q.9 Let $J = \int_0^1 \frac{\sin x}{\sqrt{x}} dx$ and $K = \int_0^1 \frac{\cos x}{\sqrt{x}} dx$, then
 (A) $J > \frac{2}{3}$ and $K > 2$ (B) $J < \frac{2}{3}$ and $K < 2$
 (C) $J < \frac{2}{3}$ and $K > 2$ (D) $J > \frac{2}{3}$ and $K < 2$

- Q.10 Let $f(x^3 + 2x - 1) = x$. If $\int_{11}^{71} \frac{dx}{3(f(x))^3 - 2f(x)} = \ln\left(\frac{p}{q}\right)$, where p and q are relatively prime numbers, then $(p - q)$ equals
 (A) 13 (B) 9 (C) 7 (D) 2

[PARAGRAPH TYPE]

Q.11 to Q.14 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Paragraph for question nos. 11 & 12

Let α, β, γ and δ be 4 distinct roots of the equation $x^4 - 4x + 3 = x(x^3 - f'(1)x^2 + f''(1)x - 4) + f(1)$ and $f(x)$ is a monic polynomial of degree 3.

- Q.11 The value of $\lim_{x \rightarrow 1} \frac{(x-1)^3 - \sin^3(x-1)}{(f(x)-3)^{5/3}}$ is
 (A) $\frac{1}{2}$ (B) 2 (C) $\frac{1}{6}$ (D) $\frac{2}{3}$
- Q.12 If $\int \frac{dx}{\frac{f(x)-3}{x-1} + 4} = \frac{1}{2} g(x) + C$, where C is constant of integration and $g(3) = \frac{\pi}{4}$, then the value of $g(5) + g(7)$ is
 (A) $\frac{\pi}{2}$ (B) $\frac{3\pi}{4}$ (C) π (D) $\frac{5\pi}{4}$

Paragraph for question nos. 13 & 14

Let f be a differentiable function satisfying the relation $\ln(f(x+y)) = \ln(f(x)) + \ln(f(y)) + 2xy - 3 \quad \forall x, y \in \mathbb{R}$ and $f'(0) = 2f(0)$.

Q.13 The value of $\frac{f'(2)}{f(2)}$ is equal to

- (A) 2 (B) 3 (C) 6 (D) 7

Q.14 The value of $\int_0^1 f(x) dx + \int_{e^3}^{e^6} (-2 + \sqrt{\ln x - 2}) dx$ is equal to

- (A) e^3 (B) e^6 (C) $e^6 - e^3$ (D) $2e^3 - e^6$

[MULTIPLE CORRECT CHOICE TYPE]

Q.15 to Q.17 has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct.

Q.15 If $\int \frac{\sin\left(\frac{2(1-\sqrt{x})}{1+\sqrt{x}}\right)}{\sqrt{x}(1+\sqrt{x})^2} dx = \left\{ f\left(\frac{1-\sqrt{x}}{1+\sqrt{x}}\right) \right\}^2 + C$, then which is(are) incorrect about $y = f(x)$?

- (A) $f(x)$ is an even function (B) $f(x)$ is an increasing function
(C) $f(x)$ is periodic function (D) $f(x)$ is not bounded

Q.16 Let $\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \frac{\pi + 4x^3}{2 - \cos\left(|x| + \frac{\pi}{3}\right)} dx = \frac{4ab}{\sqrt{3}}$, then which of the following options can be true?

- (A) $a = \pi$; $b = \tan^{-1} \frac{1}{2}$ (B) $a = \frac{\pi}{2}$; $b = \cot^{-1} 2$
(C) $a = \frac{\pi}{2}$; $b = \tan^{-1} \frac{1}{2}$ (D) $a = \pi$; $b = \cot^{-1} 2$

Q.17 If $\int \frac{dx}{(1+\sqrt{x})^{2010}} = \frac{2}{\alpha(1+\sqrt{x})^\alpha} - \frac{2}{\beta(1+\sqrt{x})^\beta} + C$

where C is constant of integration and $\alpha, \beta > 0$, then

- (A) $|\alpha - \beta| = 1$
(B) $\frac{\beta+2}{(2010)^2} = \frac{1}{\alpha+1}$
(C) $\beta, \alpha, 2010$ (taken in that order) are in arithmetic progression
(D) $\alpha + 1 = \beta + 1 = 2010$

[MATRIX TYPE]

Q.18 has four statements (A, B, C, D) given in **Column-I** and four statements (P, Q, R, S) given in **Column-II**. Any given statement in **Column-I** can have correct matching with one or more statement(s) given in **Column-II**.

Q.18 If $\int \frac{(\sin x + \cos x) + x(\sin x - \cos x)}{(1 + \sin 2x)} dx = \frac{f(x)}{g(x)} + c$, where $f(x)$ is an algebraic linear polynomial with

leading coefficient unity and $g(x)$ is trigonometric function.

[Note : $[k]$ denotes greatest integer less than or equal to k .]

Match the entries in Column-I with those in Column-II.

Column I	Column II
(A) $\int_{-2}^2 [f(x)] dx$ equals	(P) 1
(B) $\lim_{x \rightarrow \frac{-\pi}{4}} \left(\frac{g(x)}{x + \frac{\pi}{4}} \right)$ equals λ , then λ^2 equals	(Q) 2
(C) $\int_{\pi/4}^{3\pi/4} g(x) dx$ equals λ , then λ^2 greater than	(R) 3
(D) Number of solutions of the equation $(g(\pi x))^2 = f(x)$ equals	(S) -2

[INTEGER TYPE]

Q.19 & Q.20 are "Integer Type" questions. (The answer to each of the questions are upto 4 digits)

Q.19 If $f'(x) = \frac{5x^3 - 6}{\sqrt{x^7 - 3x^4 + 4x^2}}$ and $f(2) = \sqrt{6}$, then find the value of $\int_0^2 x f(x) dx$.

Q.20 If $f(x)$ is twice differentiable function from $\mathbb{R} \rightarrow \mathbb{R}$ such that $t^2 f(x) - 2t f'(x) + f''(x) = 0$ has two equal

values of t for all x and $f(0) = 1, f'(0) = 2$, then find the value of $\left[\lim_{x \rightarrow 0} \left(\frac{f(x) - 1}{2x} \right)^{\frac{1}{x}} \right]$.

[Note : $[k]$ denotes greatest integer function.]

ANSWER KEY

Q.1 A	Q.2 D	Q.3 A	Q.4 D	Q.5 B
Q.6 D	Q.7 D	Q.8 D	Q.9 B	Q.10 A
Q.11 A	Q.12 B	Q.13 C	Q.14 A	Q.15 BD
Q.16 AD	Q.17 ABC	Q.18 (A) S ; (B) Q ; (C) P, S ; (D) R		
Q.19 2	Q.20 2			